

Reliability Enhancement of the Croc SoC via Byte-Level Hardware SECDED

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1 Introduction and Project Motivation

For our VLSI2 project, we chose to focus on the reliability of the *Croc* SoC rather than its raw performance. Given its architectural positioning as a small and efficient chip, we believe that enhancing its robustness makes it significantly more competitive for specialized applications. By integrating an error correction algorithm into the memory subsystem, the chip becomes suitable for microcontrollers operating in space missions or high-radiation environments, where ionizing particles frequently cause unintentional bit flips.

2 Hardware SECDED and the Inverted Hsiao Algorithm

We implemented a **Single Error Correction, Double Error Detection (SECDED)** mechanism directly in hardware. We opted for the **Inverted Hsiao** algorithm because it results in a more balanced critical path and a lower total gate count, significantly reducing propagation latency compared to the irregular trees typically generated by Hamming implementations.

The choice of a **hardware implementation** of the logic tree, as opposed to a software-based approach, was driven by performance and efficiency requirements. A hardware codec allows for combinatorial, single-cycle error detection and correction. This ensures that the memory access maintains its timing requirements without stalling the processor. A software implementation would have introduced a massive computational overhead for every load and store operation, drastically reducing the effective throughput of the SoC.

3 Byte-Level Architecture and Integration

To natively support RISC-V byte-aligned operations, we designed a **byte-level SECDED** where each 8-bit byte is independently encoded into 13 bits (8 data bits + 5 parity bits). To accommodate these parity bits, the system memory was upgraded from baseline 32-bit structures to 64-bit wide SRAM blocks.

Although area-intensive, this approach works for our specific constraints. The *Croc* chip is currently **pad-limited**: its physical dimensions are constrained by the large number of I/O pads rather than the area occupied by standard cells or memory. Consequently, we can afford the additional logic and memory width to obtain better latency and maintain design simplicity. This architectural choice prioritizes deterministic performance and high throughput by intentionally bypassing the latency penalties inherent to **Read-Modify-Write (RMW)** cycles as in the case of conventional ECC. We are fully aware of the 12-bit overhead per memory bank, however, in this initial stage of the design, these spare bits are intentionally left unutilized during both read and write operations, as highlighted in the block diagram.

4 Implementation Details

We instantiated one hardware SECDED encoder and decoder unit for each individual byte of the original 32-bit data word. Since the architecture features two parallel SRAM blocks, a total of eight codec units were deployed. This hardware layer transparently intercepts the memory interface signals. Consequently, the remaining processing core and peripheral subsystems operate without modification, receiving and loading identical data patterns as before.

Uncorrectable double-bit errors are aggregated across all bytes via a logical OR and routed to the processor's `interrupts_o` port. This triggers an immediate hardware interrupt, allowing the RISC-V core to trap the fatal fault and prevent silent data corruption.

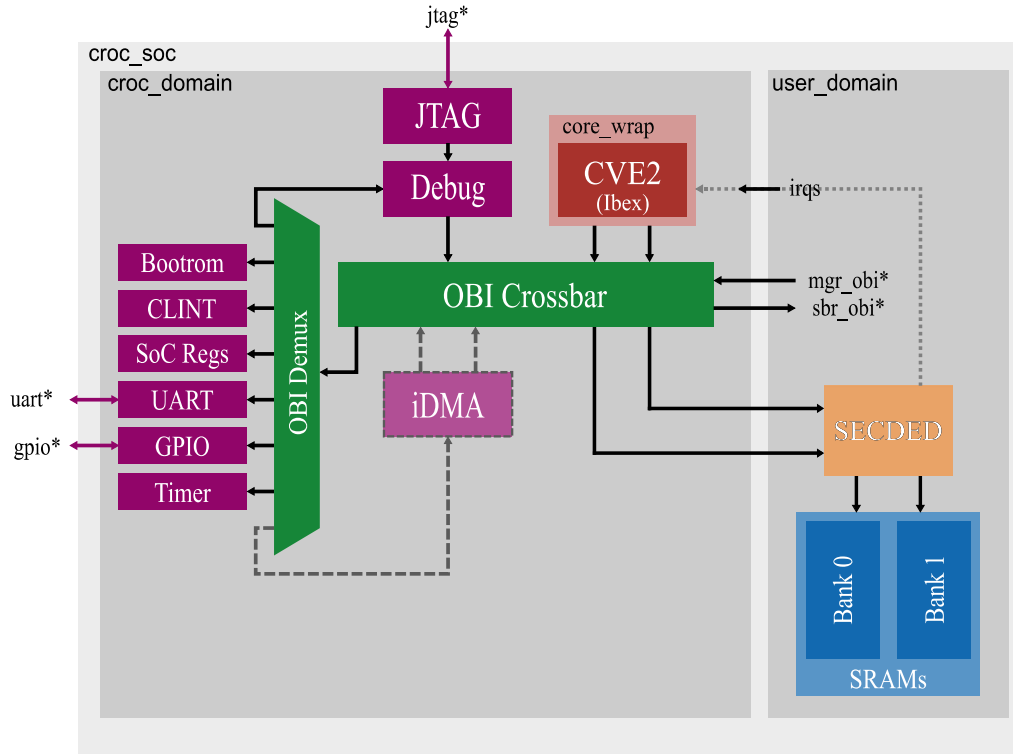


Figure 1: Simplified Block Diagram from a Top View

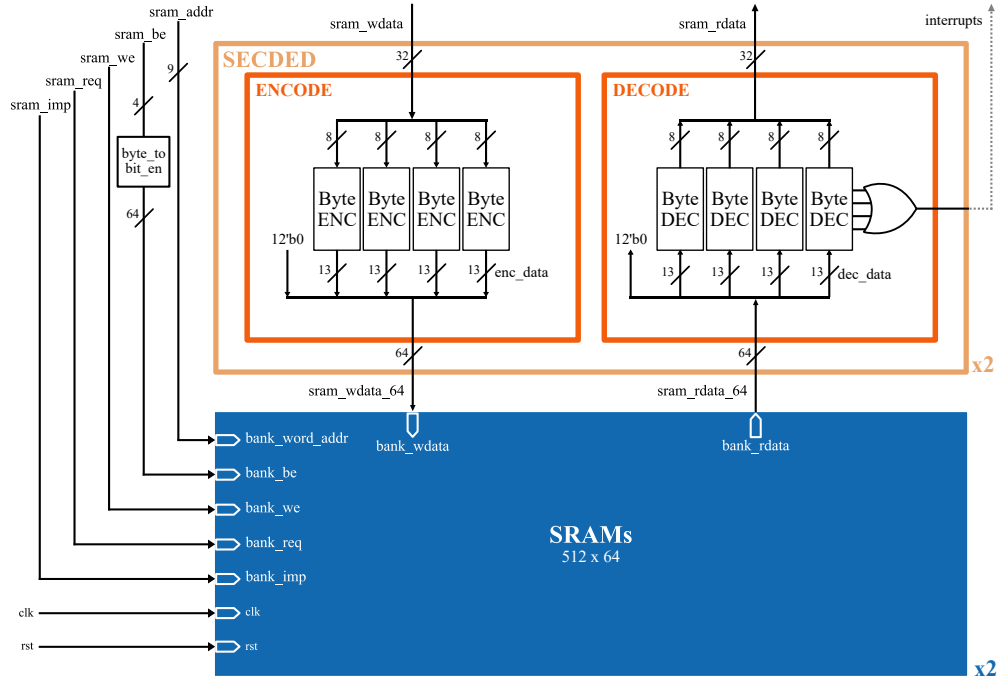


Figure 2: Simplified Block Diagram zoomed on the ENC/DEC block and SRAMs connections